

Helix Thready — HarmonyOS (ArkTS)

Customization Spec

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Rev	Date	Author	Change
1	2026-07-22	swarm (design · platforms)	Initial consolidation of the scattered per-screen ArkTS one-liners into one design contract: navigation idiom, system back, safe areas, system dark mode, Barrier-Free mapping, ArkUI substitutions table (derived), layered icon, motion honesty, i18n; minted <code>THREADY-DES-PLAT-03/-04</code>

Honest status header. The HarmonyOS client is a `helix_shims` **skeleton** `[GAP: 8.5]` — verified as such in `platform-map §2` (“Only reachable via native clients + `helix_shims` per ground truth; `helix_shims` interface not inspected — **ASSUMED** entire column”). **Nothing in this file is build-ready or claimed to work.** This spec is the **design contract the client builds against** when `[GAP: 8.5]` closes; every ArkUI API name below is **ASSUMED** until verified against real HarmonyOS SDK docs and the `helix_shims` contract — tracked as `[OPEN: THREADY-DES-PLAT-03]` (companion to the inherited `[OPEN: THREADY-DES-LIB-03]`).

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1. Position & sources consolidated

Before this file, HarmonyOS guidance lived as one-liners across: the per-screen notes panels of all 11 mobile artifacts (mobile README §2 catalogue), the wireframes §6.1 platform table (a11y / theme / font-scale row), platform-map §3 AR column + §4 notes (toast/tooltip rules), brand-assets §5.1 / icon-export-matrix §3.7 (layered icon), and design-system §7 (fan-out row). This file consolidates them and **extends** where the review found holes; the sources stay authoritative for their own scope (screens for per-screen chrome, platform-map for per-component cells).

Typography for this platform is specified in [typography-substitution.md](#) (bundling via the ArkTS font-registration API — ASSUMED; fallback HarmonyOS Sans; `fp` scale mapping) and is not repeated here.

2. Navigation idiom (5-tab IA → ArkUI)

The wireframes §6 IA `[VERIFIED structure]` — five bottom tabs (Home / Channels / Search / Assets / More) with per-tab stacks — realizes in ArkUI as (all mappings ASSUMED):

- **Root:** ArkUI `Tabs` with a bottom `TabBar` mirroring the five-tab IA (the home-feed artifact’s note already fixes this: “bottom `Tabs` bar mirrors this five-tab IA”).
- **Per-tab stacks:** ArkUI `Navigation` + `NavPathStack` per tab, so each tab keeps its own back stack (Channels → threads → post → media), matching the Android/iOS behavior in the chrome-toggle spec (mobile README §1).
- **Sheets:** the Add-Channel wizard is a `bindSheet` panel (add-channel artifact note `[VERIFIED as the design intent, ASSUMED as API]`) — same step gating and data-preservation rules as the §3.4 wizard model.

Per-screen realization, consolidated from the artifact notes panels (each cell ASSUMED, `[GAP: 8.5]`):

Screen (artifact)	ArkUI realization (consolidated from the notes panels)
Home feed	<code>Tab</code> s / <code>List</code> / <code>Column</code> declarative components; live region for processing events
Channels	<code>List</code> with swipe-action items; Add-channel via bottom sheet; system back to Home
Add-Channel	<code>bindSheet</code> panel; back dismisses, preserving entered data
Channel threads	<code>List</code> + swipe-action items; system back returns to Channels list
Post detail	pipeline pane as ArkUI <code>Progress</code> rows; Barrier-Free announces per-step status
Search	ArkUI <code>Search</code> component + chips row
Assets grid	<code>WaterFlow</code> realizes the masonry; Barrier-Free labels mirror tile semantics
Media viewer	<code>Video</code> / <code>Image</code> components + <code>AVSession</code> for background audio; system back closes
Account / Settings	same card stack as Android/iOS; three-state theme choice (§5)
Notifications	system push via the platform notification kit (<code>[OPEN: THREADY-DES-15]</code> surface)

3. System back behavior

`[DEFAULT – adjustable]`, extending the verified one-liners (“system back gesture/key maps to the same back stack” — home-feed note):

1. Back pops the **current tab’s** stack first (media viewer → post → threads → channels).
2. An open sheet/dialog consumes back as *dismiss* — and the Add-Channel sheet **preserves entered data** on dismissal (parity with the §3.4 wizard model: “Back / Esc never loses input” `[VERIFIED rule – add-channel artifact]`).
3. At a non-Home tab root, back returns to the Home tab; at Home root, back follows the system default (background the app). This two-step rule mirrors Android predictive-back convention and is a **proposal** — confirm against HarmonyOS HIG at integration (`[OPEN: THREADY-DES-PLAT-03]`).
4. Destructive flows never bind back as *confirm* — back always cancels.

4. Safe areas & notch

[DEFAULT — adjustable], ASSUMED APIs: content respects system avoid areas (status bar, punch-hole/notch, navigation/gesture area) via ArkUI safe-area handling (`expandSafeArea` only for full-bleed surfaces — the media viewer background — with controls inset to the safe area). The bottom tab bar sits above the gesture indicator area. The mobile artifacts' bezel/status-bar cosmetics are illustrative chrome, not token-governed (mobile README §3 [VERIFIED convention]); the HarmonyOS status-bar treatment follows the same rule.

5. Dark mode via system

Verified design intent (settings artifact note): “dark mode follows system + explicit override, same three-state choice”. Realization (ASSUMED APIs):

- The three-state Light/System/Dark control writes the app color-mode preference; **System** follows the OS dark mode, matching web mechanism semantics (theming §2) without reusing its CSS mechanics.
- Colors come **only** from the generated token binding (design-system §7 token bridge — the ArkTS output of the same codegen that emits `ThreadyColors` /Lipgloss; generator itself [OPEN: THREADY-DES-LIB-04], does not exist yet). Dark values are the verified Thready dark theme tokens; no ArkUI default palette may leak into token-governed surfaces.
- Per-Account white-label re-tints via the same bridge from `GET /v1/accounts/{id}/branding` (theming §8 mechanism [VERIFIED as stated mechanism]); semantic colors are never overridden (`--danger` rule, design-system §6.2).

6. Barrier-Free accessibility mapping

The a11y contract (design-system §6 [VERIFIED]) realized on the Barrier-Free kit (wireframes §6.1 row [VERIFIED table]); every ArkUI attribute name ASSUMED):

Contract point (design-system §6)	HarmonyOS realization
WCAG 2.2 AA contrast	Carried by the tokens themselves (AA-pinned values) — no platform work beyond <i>using</i> the bridge
Brand ≠ text; danger ≠ brand	Same token discipline; <code>Progress</code> /chip states use semantic tokens only
Focus visible everywhere	Focus-highlight on every interactive component (ArkUI focus states); keyboard/remote focus order follows reading order
Keyboard + screen reader	Barrier-Free screen reading: <code>accessibilityText</code> / <code>accessibilityDescription</code> on every interactive; groups via <code>accessibilityGroup</code> ; per-step pipeline status announced on change (post-detail note <code>[VERIFIED intent]</code>)
Reduced motion	Honor the system animation-reduction setting; freeze to poster frames per motion §5 (see §9)
Localized, direction-agnostic	§10; string resources per locale

Long-press tooltips replace hover hints (platform-map §4 `[VERIFIED rule]` : “Compose/Flutter/ ArkTS show tooltips on long-press — hints that matter must also exist as visible helper text”).

7. ArkUI component substitutions

Consolidated from platform-map §3 (AR column) + §4 notes; the three explicitly noted mappings first, then the derived component-group mappings. **Every row is ASSUMED** (`[GAP: 8.5]` , `[OPEN: THREADY-DES-PLAT-03]`) — the AR column has zero VERIFIED cells (platform-map §5).

Component group	ArkUI substitution	Source
Toast / alert	<code>promptAction.showToast</code> ; never the only channel for a critical error (also inline)	platform-map §4 (explicit)
Progress bar	<code>Progress</code> (det./indet.); failed+retry keeps <code>--danger</code> + focusable retry	platform-map §3/§4 (explicit)
Tooltip	long-press <code>Popup</code> hint + visible helper text for load-bearing hints	platform-map §4 (explicit)
Buttons (primary/secondary/ghost/destructive)	<code>Button</code> with token-bridge styles; destructive always <code>--danger</code>	derived from §3 AR column
Input / textarea	<code>TextInput</code> / <code>TextArea</code> + form hint slot for label/hint/error triple (<code>isError</code> -equivalent semantics travel with the message text)	derived
Select	<code>Select</code>	derived
Checkbox / radio / switch	<code>Checkbox</code> / <code>Radio</code> / <code>Toggle(ToggleType.Switch)</code> — accent track from the token bridge (platform-map §4)	derived
Date	<code>DatePickerDialog</code> (native picker wins on mobile — platform-map §4 rule)	derived
Card / stat card	<code>Column</code> + border/tokens; composites assembled, not forked	derived
Table → cards	On phones the table collapses to thread-row cards (platform-map §4 rule); <code>List</code> + header where a table survives	derived
Badge / chips	<code>Badge</code> ; hashtag <code>Chip</code> (direct = brand fill, indirect = outline + <code>--muted</code> + non-color marker); processing chip = chip + progress, five states + <code>[r]</code> -equivalent retry (component-library §7 parity contract)	derived
Dialog / modal	<code>CustomDialog</code> ; destructive action never default-focused	derived
Spinner / helix motif	<code>LoadingProgress</code> ; helix-motif spinner via <code>Canvas</code> drawing	derived
Tabs / sidebar / topbar		derived

Component group	ArkUI substitution	Source
	<code>Tabs</code> / <code>SideBarContainer</code> / <code>Navigation</code> title bar	
Skeleton	opacity-pulse rows, only after ~150 ms (no flash — platform-map §4 rule)	derived
Empty / error state	centered composites; error page + idempotent retry	derived
Avatar / pagination / breadcrumbs / link	<code>Image</code> + circle text / button row / text row / <code>Span</code> + click	derived
Theme toggle / language picker	dark-mode config three-state (§5) / locale select (§10)	derived

8. Layered icon integration

Grounded in icon-export-matrix §3.7 [VERIFIED — assets produced by the live generate-raster.sh run] :

- `harmonyos/foreground.png` (216) + `harmonyos/background.png` (216) referenced from `resources/base/media/layered_image.json` (`foreground` / `background`); the solid/branded background layer is supplied at integration (brand-assets §5.1).
- `harmonyos/appgallery-1024.png` for the AppGallery listing.
- The **assets exist now; no HarmonyOS launcher is claimed** — wiring them into an actual `.hap` is [GAP: 8.5] work. The launcher icon carries the helix element, no letters [VERIFIED — brand-assets] .

9. Motion

What ground truth actually says — checked, not assumed: motion.md §6 names **exactly three** Lottie runtimes (`lottie-web` , `lottie-compose` , `lottie-ios`); neither motion.md nor platform-map names **any** HarmonyOS Lottie runtime. Therefore:

- A community `@ohos/lottie` player is believed to exist in the OHOS ecosystem, but it is **unverified here — ASSUMED**, and adopting it (vs. reimplementing the small animation set in ArkUI `animateTo`) is [OPEN: THREADY-DES-PLAT-04] . THREADY-MOT-01's verify-before-ship rule extends to any HarmonyOS runtime.
- **Guaranteed floor (no Lottie needed):** the five static `reducedMotionFallback` SVGs/posters (motion §5 [VERIFIED delivered]) render the loading/success/error/pulse/sync states; simple state transitions use ArkUI animation with the token durations `--motion-fast 150ms` / `--motion-base 200ms` on `--ease-standard cubic-bezier(0.2,0,0,1)` [VERIFIED values] .

- Reduced motion: the system animation-reduction setting gates all of it, freezing to the exact poster frames of motion §5 (equivalent-runtime rule: “both paths render the same state by construction” `[VERIFIED – motion §5]`).

10. i18n

- Locales: `en` / `ru` / `sr-Cyrl` `[OPERATOR §12]` , following the **system locale** by default with the same in-app language picker semantics as every surface (component-library §4 `I18nService` key parity — including `footer.made` / `footer.by` / `ally.love` for the locked attribution footer `[VERIFIED keys]`).
- String delivery via HarmonyOS resource qualifiers per locale (ASSUMED mechanism); the string *keys* stay the shared dictionary so translations are written once.
- Cyrillic rendering per *typography-substitution §5* — HarmonyOS Sans coverage is an unverified expectation `[OPEN: THREADY-DES-PLAT-02]` .
- The token layer is direction-agnostic (design-system §6.6); all three locales are LTR.

11. Open items

- `[OPEN: THREADY-DES-PLAT-03]` — **verify every ArkUI mapping in this file** (Navigation/Tabs/NavPathStack, `bindSheet` , `WaterFlow` , `AVSession` , `promptAction` , `font.registerFont` , Barrier-Free attributes, safe-area APIs, back-at-tab-root rule) against the current HarmonyOS SDK/HIG **and** the `helix_shims` contract once it exists. Companion to the inherited `[OPEN: THREADY-DES-LIB-03]` .
 - `[OPEN: THREADY-DES-PLAT-04]` — **HarmonyOS motion runtime**: verify `@ohos/lottie` (or choose ArkUI reimplementations); until then only the §9 guaranteed floor may be claimed.
 - Inherited: `[GAP: 8.5]` (client skeleton — release gate, wireframes §6.2), `[OPEN: THREADY-DES-LIB-03]` (`helix_shims` uninspected), `[OPEN: THREADY-DES-LIB-04]` (token-bridge codegen does not exist), `[OPEN: THREADY-DES-15]` (notifications surface), `[OPEN: THREADY-DES-PLAT-01/-02/-08]` (typography, via *typography-substitution.md*).
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